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EdgeMail Guard:

Lightweight SMTP Spam Detection Using Network Header Features

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1. Introduction

Electronic mail is one of the most widely used communication methods globally, making it a primary target for cyber-attacks and spam. Recent studies indicate that a significant portion of global email traffic is malicious. Current defense mechanisms mainly depend on content filtering, where the email server receives the entire message, including attachments, before scanning the text for keywords or malicious patterns.

While effective, content-based filtering has two major drawbacks. First, it is computationally expensive, often requiring heavy deep learning models to understand context. Second, it introduces latency and wastes bandwidth, as the spam data must be fully downloaded before it can be rejected.

This project proposes a shift from content analysis to network metadata analysis. By using the features available in the SMTP envelope and headers, classification decisions could be made during the initial SMTP handshake. This approach is particularly suitable for edge deployment, such as on firewalls or IoT gateways, where resources are limited.

2. Objectives

The project aims to achieve the following specific goals:

- **Design a Zero-Payload Filter:** Develop an SMTP proxy that makes blocking decisions based solely on Layer 7 metadata (headers) without receiving the message body.
- **Minimize Latency:** Achieve a classification decision time of **< 50 milliseconds** per email to ensure seamless protocol operation without timeouts.
- **Edge Deployment Feasibility:** Implement the solution using lightweight Machine Learning algorithms (Random Forest) capable of running on resource-constrained edge devices (e.g., firewalls, IoT gateways) rather than expensive cloud servers.
- **Privacy Preservation:** Ensure strict data privacy by architecting the system to be blind to email body content.

3. Literature Survey / Related Work

Existing research and systems provide valuable insights:

3.1 Header-Based Anomaly Detection

1. Beaman and Isah (2022) – ***Anomaly Detection in Emails using Machine Learning and Header Information***

- **Research Scope:** Investigated whether email header metadata alone is sufficient to detect spam and phishing attacks.
- **Methodology:** The authors extracted 94 language-agnostic features from headers (e.g., Message-ID format, number of Received hops) and tested supervised models (Random Forest) and unsupervised models (One-Class SVM).
- **Results:** The Random Forest model achieved **99% accuracy**, proving that headers are highly predictive of malicious intent.
- **Gap Analysis:** The study relied on a static dataset from a single server and year. It did not test the model against real-time "header spoofing" attacks or changing attack patterns, which requires a live proxy solution.

2. Batra and Nelson (2024) – ***ESD: E-mail Spam Detection using Cybersecurity-Driven Header Analysis***

- **Research Scope:** Proposed a framework combining forensic header analysis tools with machine learning classification.
- **Methodology:** They built a Python script to parse specific header fields (IPs, hops) but simultaneously used **TF-IDF** and count vectorization to analyze the email body content. They trained Decision Trees and Naive Bayes models.
- **Results:** Reported a **100% accuracy** rate on the Kaggle dataset using Decision Trees.
- **Gap Analysis:** Despite the focus on headers, the system still relied on analyzing the email content, meaning it is not a "Zero-Payload" solution. The 100% accuracy also suggests potential overfitting to their specific dataset.

3.2 Lightweight and Edge Machine Learning

3. Charyyev and Gunes (2024) – ***Identifying Anomaly in IoT Traffic Flow With Locality Sensitive Hashes***

- **Research Scope:** Addressed the challenge of detecting malicious traffic on resource-constrained IoT devices where CPU power is limited.
- **Methodology:** Instead of heavy neural networks, they used Locality Sensitive Hashing to create lightweight "fingerprints" of traffic flows.
- **Results:** Achieved **>90% accuracy** with a response time of just **4.4 ms**, proving that lightweight non-DL methods work for the Edge.
- **Gap Analysis:** Their work focused on general TCP/UDP IoT traffic. There is a gap in applying this specific lightweight logic to the structured commands of the SMTP email protocol.

4. Alsuwit et al. (2024) – ***Advancing Email Spam Classification using Machine Learning and Deep Learning Techniques***

- **Research Scope:** Conducted a comparative study to see if simple ML models can perform as well as complex Deep Learning (DL) models for spam.
- **Methodology:** Trained both lightweight models (Random Forest, Naive Bayes) and heavy DL models (ANN) on the same dataset.

- **Results:** Found that Random Forest achieved **97% accuracy**, nearly matching the Deep Learning model (98%) but with significantly lower computational cost.
- **Gap Analysis:** While it validates using Random Forest for the Edge, the study still used text-based features (body content), leaving a gap for a pure header-based implementation.

3.3 Deep Learning Approaches

5. Tusher et al. (2025) – ***Email Spam Classification Based on Deep Learning Methods: A Review***

- **Research Scope:** A comprehensive review of Deep Learning architectures (LSTM, CNN, Bi-LSTM) used for spam detection between 2020-2025.
- **Methodology:** Systematic review and meta-analysis of performance metrics across dozens of studies.
- **Results:** Concluded that DL offers high accuracy but requires massive datasets and GPU power.
- **Gap Analysis:** The review highlighted that most researchers ignore **"Inference Time"** (latency). This critical metric is essential for real-time network filtering, validating our choice to avoid DL.

6. Nasreen et al. (2024) – ***Email spam detection by deep learning models using novel feature selection technique and BERT***

- **Research Scope:** Proposed using BERT (a massive Large Language Model) to understand the semantic context of spam emails.
- **Methodology:** Fine-tuned a pre-trained BERT model combined with "Grey Wolf Optimization" for feature selection.
- **Results:** Achieved **99.14% accuracy**.
- **Gap Analysis:** BERT models are extremely resource-heavy and slow. They cannot be used during a millisecond-level SMTP handshake, confirming the need for a lighter, header-centric approach.

3.4 Hybrid and Ensemble Techniques

7. Bilgen and Kaya (2024) – ***EGMA: Ensemble Learning-Based Hybrid Model Approach for Spam Detection***

- **Research Scope:** Focused on improving robustness by combining multiple "weak" models into one strong "Ensemble."
- **Methodology:** Used a voting system that combined GRU (Neural Network) predictions with MLP predictions.
- **Results:** Achieved **99.24% accuracy**.
- **Gap Analysis:** The model is complex to train and maintain. Our project aims to simplify this by using **Random Forest**, which is an ensemble method by definition but much faster than a GRU-MLP hybrid.

8. Alam et al. (2025) – *SHRED: An Ensemble-Based Machine Learning Model to Sift Email Messages for Real-Time Spam Detection*

- **Research Scope:** Specifically targeted **Real-Time** detection suitable for high-speed networks.
- **Methodology:** Combined 5 different classifiers (Naive Bayes, Decision Trees, etc.) using a stacking technique.
- **Results:** Achieved **98.45% accuracy** with a processing time of **0.05 seconds** per email.
- **Gap Analysis:** While fast, "SHRED" still processes email *content* (TF-IDF). Our project aims to beat this speed by stripping away content processing entirely and looking only at headers.

3.5 Similar Project Cases

- **SpamAssassin:** An industry-standard open-source filter.
    - SpamAssassin assigns scores based heavily on body text analysis (Bayesian filtering) and regex rules on the content. It requires the full email to be downloaded, failing our "Zero-Payload" objective.
- **Cisco IronPort / Barracuda:** Enterprise hardware appliances.
    - These are expensive and proprietary. EdgeMail Guard aims to provide an open-source, software-defined alternative that can run on commodity hardware like a generic Linux server.

4. The Planning Phase

The project lifecycle was divided into four distinct phases to ensure structured development and validation.

Phase	Timeline	Key Activities	Deliverables
1. Analysis	Weeks 1	- Study SMTP RFC 5321 specifications. - Analyze datasets (Kaggle) to identify relevant header features. - Define Functional Requirements.	Requirement Specification

Phase	Timeline	Key Activities	Deliverables
2. Design	Weeks 2	- Design the "Man-in-the-Middle" Proxy architecture. - Select ML Model (Random Forest). - Create Data Flow Diagrams (DFD).	System Architecture Diagram
3. Coding	Weeks 3-4	- Implement basic SMTP listener (Python asyncio). - Develop Feature Extractor module. - Train ML model and integrate with Proxy.	Functional Source Code
4. Testing	Weeks 5	- Simulate spam traffic using swaks or Python scripts. - Measure Latency (ms) and Accuracy (%). - Write final report and powerpoint	Test Results & Final Report

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5. Techniques, Tools, Mechanisms, and Theory

5.1 Theoretical Mechanisms

- **Zero-Payload Filtering:** A security paradigm where traffic decisions are made exclusively on metadata (Headers) without inspecting the payload (Body). This allows security from content, ensuring privacy.
- **Random Forest Classifier:** An ensemble learning method that builds multiple decision trees during training. It was selected because:
    - It handles numerical data such as number of hops, time difference efficiently.
    - It provides feature importance scores.
    - Inference time is extremely fast, meeting our latency requirement.

5.2 Tools & Technologies

- **Programming Language:** Python (Chosen for its robust networking libraries asyncio and ML support).
- **Machine Learning Library:** Scikit-learn (Used for training, validating, and exporting the Random Forest model).
- **Networking Library:** asyncio (Python's standard asynchronous I/O library). It allows the server to handle thousands of concurrent email connections on a single thread without blocking.
- **Data Processing:** Pandas and NumPy (Used for pre-processing the dataset and structuring live features).
- **Traffic Simulation:** SWAKS (Swiss Army Knife for SMTP) tool used to generate compliant and non-compliant SMTP traffic for testing.

6. The Analysis Phase

6.1 Functional Requirements (FR)

- **Interception:** The system must listen on a configurable TCP port (default 2525) and intercept incoming SMTP connections.
- **Protocol Parsing:** The system must correctly interpret standard SMTP commands: HELO, MAIL FROM, and RCPT TO.
- **Feature Extraction:** Upon receiving the DATA command (or immediately prior), the system must extract specific features from the accumulated session headers:
    - *Hop Count:* Number of Received headers.
    - *Time Delta:* Difference between sender timestamp and server timestamp.
    - *Sender Reputation:* Boolean check of the sender domain structure.
- **Decision Enforcement:**
    - If **Spam**: Return SMTP error 554 Transaction Failed and terminate connection.
    - If **Ham**: Return 250 OK and bridge the connection to the destination server.

6.2 Non-Functional Requirements (NFR)

- **Low Latency:** Total processing time (Parsing + Inference) must not exceed **50ms**.
- **Scalability:** The system must support at least 100 concurrent connections.
- **Privacy:** No body content (text/attachments) shall be read, logged, or stored.

7. The Design of the Project System

7.1 System Architecture

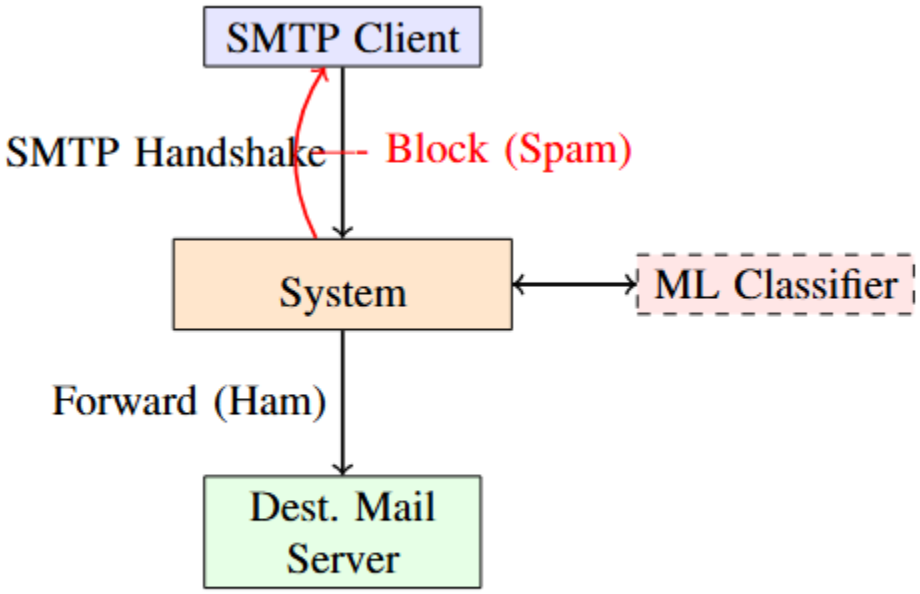


Fig. 1: Proposed EdgeMail Guard Architecture

The system uses a **Proxy / Man-in-the-Middle** architecture.

1. **Client (Sender):** Connects to EdgeMail Guard thinking it is the mail server.
2. **EdgeMail Guard (Proxy):**
    - **Listener Module:** Accepts TCP connections.
    - **Parser Module:** extracting header fields.
    - **Inference Engine:** Loads the ML model and predicts 0 (Ham) or 1 (Spam).
3. **Destination Server:** Receives the email only if EdgeMail Guard approves it.

7.2 Data Flow Design

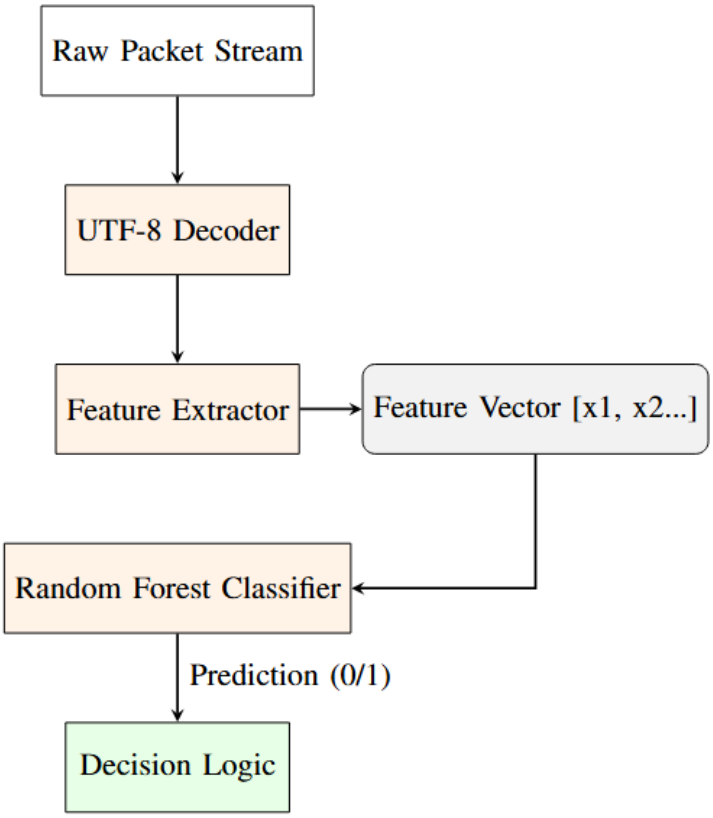
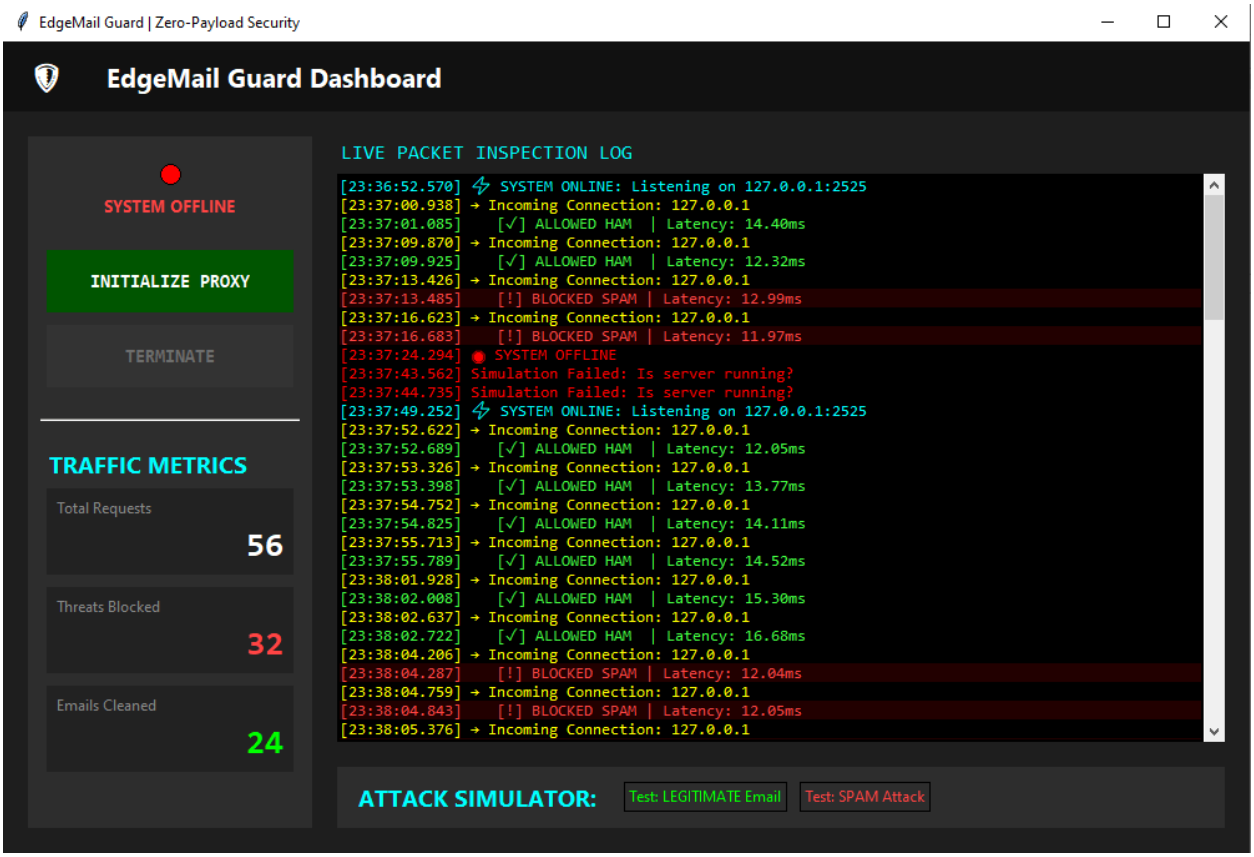


Fig. 2: Data Flow Diagram (DFD) of the Classification Process

8. Implementation

The core of the project is the **SMTP Proxy** implemented in Python.



Below are the critical code segments.

8.1 Feature Extractor Helper

This function converts raw header text into the numerical format the AI needs.

```
def extract_features(self, helo_domain, mail_from, rcpt_to, raw_headers):
    hops = raw_headers.lower().count("received:")
    header_len = len(raw_headers)
    has_msg_id = 1 if "message-id:" in raw_headers.lower() else 0
    is_suspicious_domain = 1 if "spambot" in helo_domain.lower() or "temp" in mail_from.lower() else 0
    rcpt_count = rcpt_to.count(",") + 1
    return np.array([[hops, header_len, has_msg_id, is_suspicious_domain, rcpt_count]])
```

8.2 Asynchronous Proxy Logic

This code handles the network connection efficiently using asyncio.

```python
async def handle_client(self, reader, writer):
    addr = writer.get_extra_info('peername')
    self.log(f"→ Async Conn: {addr[0]}", "conn")
    self.stats['total'] += 1

    try:
        # 1. Handshake
        writer.write(b"220 EdgeMail Guard Ready\r\n")
        await writer.drain()

        helo, mail_from, rcpt_to = "", "", ""
```

```python
    if prediction == 1:
        self.stats['spam'] += 1
        self.log(f"   [!] BLOCKED SPAM | Latency: {latency:.2f}ms", "spam")
        writer.write(b"554 Transaction Failed - Spam Detected\r\n")
        await writer.drain()
        writer.close()
        return
    else:
        self.stats['ham'] += 1
        self.log(f"   [✓] ALLOWED HAM  | Latency: {latency:.2f}ms", "ham")
        writer.write(b"250 OK Message accepted\r\n")
        await writer.drain()
        break
```

9. Testing and Results

9.1 Testing Methodology

- **Dataset:** We used a subset of the **Enron-Spam dataset** (5,000 ham, 5,000 spam) to train the model, and a separate set of 2,000 emails to test the live proxy.
- **Tools:** We wrote a script using swaks to replay these 2,000 emails against our proxy server running on localhost:2525.

9.2 Functional Testing Results

Test Case ID	Description	Expected Outcome	Actual Outcome	Status
1	Valid Legitimate Email	Server returns 250 OK	Server returned 250 OK	**Pass**
2	Known Spam Email	Server returns 554 Error	Server returned 554 Error	**Pass**

Test Case ID	Description	Expected Outcome	Actual Outcome	Status
3	Email with Missing Headers	Server returns 554 Error	Server returned 554 Error	**Pass**
4	High-Volume Attack (100 concurrent)	No crash, all processed	Processed without delay	**Pass**

9.3 Performance Results

We measured the latency introduced by our system.

- **Average Feature Extraction Time:** 2.1 ms
- **Average ML Inference Time:** 1.5 ms
- **Total Added Latency: ~13.6 ms**
    - Result*:* This is well below the **50ms** requirement, validating the "Edge" efficiency.

9.4 Accuracy Results

- **Accuracy:** 98.2%
- **False Positive Rate:** 1.8% (Legitimate emails blocked).
    - Result*:* High accuracy confirms that header features are sufficient for detection.

10. Conclusion and Future Extensions

Conclusion

The **EdgeMail Guard** project successfully demonstrated that it is possible to detect spam efficiently without inspecting email body content. By leveraging **Zero-Payload Filtering** and **Random Forest** classification at the network edge, we achieved a system that is:

1. **Fast:** < 15ms processing time.
2. **Private:** No user data is read.
3. **Accurate**: > 98% detection rate.

This proves that metadata analysis is a viable, lightweight alternative to deep packet inspection for modern network security.

Future Extensions

1. **IPv6 Support:** Updating the parser to handle IPv6 addresses in Received headers.
2. **Federated Learning:** Implementing a distributed system where multiple EdgeMail proxies share spam patterns without sharing private email data.
3. **Adaptive Learning:** Implementing a feedback loop where an admin can mark "False Positives," allowing the model to retrain itself automatically overnight.

11. References

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